Visualization-Based OSINT Intelligence Platform

Graphs, Link Charts, Timelines and Heat Maps for Evidence-Based Threat Investigation

Project Report

Cybersecurity / OSINT Threat Intelligence

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Abstract

Open threat-intelligence feeds publish thousands of malicious indicators every day, but as flat lists. The intelligence an analyst needs is in the relationships the lists do not show: shared servers, campaign time spans, and concentrations of infrastructure. This project designs and implements a visualization-based OSINT analytics platform. It collected 3 public feeds (URLhaus, ThreatFox, OpenPhish) and enriched them with DNS, IP geolocation, an independent ASN registry service and RIR address statistics. From these it built a provenance-labelled knowledge graph of 50,391 entities and 145,463 relationships. It then compared four visualization families (graphs, link charts, timelines and heat maps), assigning each the investigative questions it answers best. Each view carries safeguards against misinterpretation. The safeguards were tested quantitatively. Without hub suppression, multi-tenant platforms such as raw.githubusercontent.com merge unrelated indicators into a false 'campaign' of 13,580 entities; with suppression the largest cluster is 368. Normalising hosting countries by allocated IPv4 space keeps only 3 of the raw top-10 countries (Spearman $\rho = 0.362$). Geolocation was cross-validated against registry data (76.0% country agreement, 98.7% ASN agreement). A literature survey of 15 papers identifies the research gaps the architecture addresses.

1. Problem Definition

Community threat-intelligence feeds such as abuse.ch URLhaus and ThreatFox, and phishing feeds such as OpenPhish, are among the most widely used open sources in security operations. Each publishes a flat list of indicators of compromise (IOCs): URLs, domains, IP:port pairs and file hashes, with timestamps, tags and reporter names. The questions an investigator actually asks are relational, temporal and spatial: Which indicators share infrastructure? Starting from one suspicious IP, what else is connected? When did a campaign start and peak? Where is the infrastructure hosted? A list answers none of these directly.

Visualization closes that gap, but each chart form can also mislead. A force-directed graph suggests a relationship wherever two nodes are drawn together. A heat map of raw counts mostly shows where hosting is cheap and plentiful. A timeline of report dates is easily read as attack dates. The problem therefore has two parts: (a) build an OSINT analytics platform that turns open data into entities, relationships, timestamps, locations and threat indicators, and visualises them; (b) choose and design each visualization so that it supports the investigative question it is used for and reduces the chance of misinterpreting the evidence.

1.1 Objectives

Table 1. Project objectives

#	Objective	Measured by
O1	Collect a multi-source OSINT dataset with entities, relationships, timestamps, locations and threat indicators	3 feeds + 4 enrichment sources; SHA-256 manifest
O2	Clean, normalise and de-duplicate; extract typed entities	rejection counts by reason; entity counts
O3	Correlate across sources into a provenance-labelled knowledge graph	cross-feed overlap; infrastructure clusters
O4	Apply graph, temporal, geographic and social-network analytics	centrality, Louvain, bursts, normalised geo, SNA
O5	Evaluate the suitability of graphs, link charts, timelines and heat maps	suitability matrix (Section 5)
O6	Implement an interactive dashboard and automated reporting	Flask dashboard, HTML reports, GEXF/CSV export
O7	Reduce, and quantify, the possibility of misinterpreting evidence	before/after experiments (Section 8.3)
O8	Survey at least 10 research papers and identify research gaps	15-paper comparison (Section 2)

1.2 Investigation questions

The investigation questions were fixed before any visualization was chosen, so each view could be judged against a concrete need rather than how it looks.

Table 2. Investigation questions

ID	Investigative question	Requirement on the view
IQ1	Which indicators share infrastructure and probably belong to the same campaign?	Show connectivity and groups at a glance
IQ2	Starting from one indicator, what is connected, and on what evidence?	Incremental, explainable expansion
IQ3	When did activity start, peak and stop? Are there bursts?	Ordered time axis, volume, durations
IQ4	Where is infrastructure hosted, and which countries/ASNs are over-represented relative to size?	Spatial density, normalisation
IQ5	What is the operational (reporting) tempo across the week?	Two-dimensional time density
IQ6	Which entities broker between clusters?	Structural importance (centrality)
IQ7	How is coverage spread across the reporting community?	Social-network structure

1.3 Scope and requirements

- **In scope:** passive collection from URLhaus, ThreatFox and OpenPhish; passive enrichment through DNS, ip-api.com, the Team Cymru IP-to-ASN service and NRO RIR delegation statistics; free-text IOC extraction; correlation, analytics, visualization, reporting and export.
- **Out of scope:** active scanning or probing, visiting or downloading malicious URLs, malware execution, attribution of attacks to real people, personal social-media collection, takedown or blocking.
- **Functional requirements:** (F1) reproducible collection with a chain-of-custody manifest; (F2) no single failing source stops the pipeline; (F3) normalisation and de-duplication; (F4) typed entities and provenance-labelled edges; (F5) correlation with hub suppression; (F6) graph, temporal, geographic and SNA analytics; (F7) interactive graph, link chart, timeline and heat-map views with an evidence panel; (F8) reports and exports; (F9) full offline reproduction from a committed snapshot.
- **Non-functional requirements:** explainability (every on-screen link traceable to a source record), reproducibility, resilience, ethical collection only, and laptop-scale performance (full pipeline in minutes).
- **Intended users:** SOC and threat-intelligence analysts triaging malicious infrastructure.

2. Literature Survey

Fifteen papers were reviewed across five areas required by the brief: OSINT and intelligence analysis, graph visualization and analytics, social-network analysis, geographic visualization, and cyber-intelligence visualization and dashboards. Table 3 compares their approach, methodology, data, findings and limitations. Section 2.2 derives the research gaps that the proposed architecture addresses.

Table 3. Comparison of surveyed research

Paper	Area	Approach / visualization	Methodology & dataset	Key findings	Limitations
[L1] Pirolli & Card (2005)	Intelligence analysis	Cognitive task analysis of intelligence analysts; sensemaking loop (foraging → schema → hypothesis → presentation)	Interviews and think-aloud with analysts	Analysts spend most effort on foraging and on keeping track of evidence; leverage points for tools identified	Conceptual; no implemented system or quantitative evaluation

[L2] Stasko, Görg & Liu (2008) Jigsaw	Investigative visual analytics	Entity extraction from documents + linked list, graph, calendar and scatter views	System design; user study with analysts on a fictional intelligence case	Multiple coordinated views of entity connections help analysts find plots faster	Text documents only; entity extraction errors propagate; small user study
[L3] Pastor-Galindo et al. (2020)	OSINT survey	Survey of OSINT sources, tools, techniques and applications	Systematic literature review	OSINT is under-exploited; integration of sources and analysis is the main challenge	No implementation; visualization treated only briefly
[L4] Böhm & Lolagar (2021)	OSINT ethics and law	Legal and ethical framework for OSINT	Legal analysis (GDPR, data protection)	Public does not mean unrestricted; purpose limitation and minimisation apply	European focus; no technical controls proposed
[L5] Herman, Melançon & Marshall (2000)	Graph visualization	Survey of graph layout, navigation and clustering techniques	Literature survey	Large graphs need clustering, filtering and focus+context navigation	Pre-dates modern web tools; no task-based evaluation
[L6] Ghoniem, Fekete & Castagliola (2004)	Graph readability	Controlled comparison of node-link diagrams vs adjacency matrices	User study, 36 participants, 7 tasks on random graphs	Node-link diagrams read better for small, sparse graphs and path tasks; matrices win for large or dense graphs	Synthetic graphs; no domain semantics
[L7] Blondel et al. (2008) Louvain	Graph analytics	Greedy modularity optimisation for community detection	Algorithm + evaluation on networks up to 118 M nodes	Fast, high-modularity communities at very large scale	Resolution limit; non-deterministic; communities are not ground truth
[L8] Krebs (2002)	Social-network analysis	SNA of the 9/11 hijacker network built from public news sources	Network reconstruction from open sources; centrality	Key brokers identifiable from public data; covert networks are sparse	Retrospective; incomplete, noisy data; single case
[L9] Xu & Chen (2005)	Criminal network analysis	SNA + visualization for criminal networks (CrimeNet Explorer)	Concept-space linkage, hierarchical clustering, blockmodelling on police records	Network visualisation helps identify subgroups and central members	Police records, not open data; link reliability not modelled
[L10] Chainey, Tompson & Uhlig (2008)	Geographic visualization	Comparison of hotspot mapping methods (KDE, grids, ellipses, thematic)	Prediction accuracy index on UK crime data	Kernel density estimation predicts future hotspots best; method choice changes the conclusion	Physical crime, not network infrastructure
[L11] Poese et al. (2011)	Geolocation reliability	Accuracy study of commercial IP geolocation databases	Comparison against ground truth from a large ISP	Country-level geolocation is mostly right, city-level often wrong; databases disagree	Single ISP ground truth; databases change over time
[L12] Shiravi, Shiravi & Ghorbani (2012)	Security visualization	Survey and taxonomy of network-security visualization systems	Classification by data source and use case	Most systems target one data type; evaluation is rare and informal	Pre-dates threat-intel feeds; few OSINT systems
[L13] Noel et al. (2016) CyGraph	Cyber graph analytics	Graph-based knowledge model and visualization of network/attack data	Property graph (Neo4j) + query language; case studies	Unified graph enables attack-path and mission-impact queries	Internal network data; no open-source feeds; limited evaluation
[L14] Wagner et al. (2016) MISP	Threat-intel platforms	Design of a collaborative IOC sharing platform with correlation	System design and operational deployment	Automatic correlation of shared attributes speeds up analysis	Correlation over shared attributes produces noisy links; basic visualisation
[L15] Correll & Heer (2017)	Misleading visualization	Catalogue of how charts deceive (black-hat visualization)	Analysis of deceptive design patterns	Aggregation, scaling and omitted context mislead even honest readers	Conceptual; no domain-specific safeguards

2.1 Comparison of visualization approaches

The surveyed work uses four families of visualization. Node-link graphs [L2, L5, L9, L13] dominate investigative and cyber work because they show connectivity directly. [L6] shows, however, that they lose readability as size and density grow, and [L5] recommends clustering and filtering. Link charts, meaning incremental ego-network expansion, are the analyst-facing form of the same model; Jigsaw [L2] and MISP [L14] both build on it. Timelines and calendar views [L2] support sequence and duration reasoning. Heat maps support density reasoning in space [L10] and in time. [L10] and [L11] both show that the chosen method and the underlying data quality change what the map appears to say. Social-network analysis [L8, L9] contributes centrality and subgroup measures that turn a picture into ranked evidence. Louvain community detection [L7] is the standard scalable way to find those subgroups.

2.2 Research gaps addressed by this project

- **G1 – Evidence provenance in the view.** Graph-based systems [L2, L9, L13, L14] draw links but rarely let the analyst see, on the link itself, which source record created it, by what method and with what confidence. Pirolli & Card [L1] identify keeping track of evidence as a key leverage point. → Every edge in this platform carries source, record ID, time, method and confidence, shown in an evidence panel.
- **G2 – False links through shared infrastructure.** Attribute-based correlation [L14] and large-graph analytics [L13] connect everything that shares an attribute. In open feeds, CDNs, code-hosting and paste platforms are shared by thousands of unrelated actors. The effect is noted informally but rarely measured. → Hub suppression, quantified before and after (Section 8.3).
- **G3 – Normalisation and uncertainty in geographic views.** Hotspot work [L10] shows that method changes the conclusion, and [L11] shows geolocation error. Cyber-threat maps nevertheless typically plot raw counts. → Distinct-IP weighting, per-million-IPv4 normalisation, and cross-validation of geolocation against registry data.
- **G4 – Temporal semantics.** Timelines in investigative tools [L2] do not distinguish event time from report time. Feeds record when something was reported, not when it was active. → Explicit report-time labelling; observation-only timestamps excluded.
- **G5 – Integrated, reproducible OSINT visualization over public data.** Surveys [L3, L12] note that most systems address one data type or view, and that evaluation is rare [L12]. → One provenance-labelled model feeds four coordinated views, built from a hash-verified public snapshot that anyone can re-run.
- **G6 – Ethics built into the tool.** [L4] sets out the legal and ethical principles, but technical controls are rarely described. → Pseudonymised reporters, defanged reports, passive-only collection.

3. OSINT Data Collection and Source Identification

The dataset had to contain entities, relationships, timestamps, locations and threat indicators. No single open source provides all five, so three primary feeds were combined with four enrichment sources. Table 4 lists every source, what it contributes, how it is accessed and how failure is handled.

Table 4. Open sources used

Source	Type	Contributes	Access method	Failure handling
abuse.ch URLhaus	Threat feed (malware URLs)	URLs, hosts, timestamps, status, tags, reporters	CSV export (recent, 30 days), no key	Retry ×3; marked failed in manifest; others continue
abuse.ch ThreatFox	Threat feed (IOCs)	IP:port, domains, URLs, hashes, explicit malware family, confidence	CSV export (recent), no key	As above

OpenPhish Community	Phishing feed	Phishing URLs (no timestamps)	Plain-text feed	As above; retrieval time marked as observation time
DNS (dnspython)	Enrichment	host → IP (A records), NXDOMAIN status	Standard resolver queries, 32 threads	Timeout/NXDOMAIN stored as status
ip-api.com	Enrichment	Geolocation, ASN, ISP, hosting flag	Batch API (100 IPs/request, ≤15 req/min)	Rate-limit headers honoured; failed batch logged
Team Cymru IP-to-ASN	Enrichment (independent)	ASN, BGP prefix, registry country	DNS TXT queries (origin.asn.cymru.com)	Missing value = no comparison
NRO RIR delegation stats	Reference data	IPv4 addresses allocated per country	Combined RIR file (57 MB, summarised)	Normalisation disabled if missing

3.1 Collection methodology and chain of custody

Every download is streamed to disk, hashed with SHA-256 and recorded in a manifest together with the source URL, UTC retrieval time, size and licence (Table 5). The download was then frozen as a gzip-compressed snapshot in data/snapshot/. Running the pipeline with OFFLINE=1 restores the snapshot, verifies every hash, and reproduces every number in this report without a single network request. Enrichment results are cached in SQLite and exported to the snapshot for the same reason.

Table 5. Chain-of-custody manifest of the analysed snapshot

Source	Retrieved (UTC)	Bytes	SHA-256 (first 24 hex)
urlhaus	2026-10-19T17:13:43+00:00	7,869,800	bc168ff83c81f1424c74424c
threatfox	2026-10-19T17:13:44+00:00	1,543,610	a435ade364277275c8df79d6
openphish	2026-10-19T17:13:47+00:00	14,307	0447401a73d47149cef29e04

3.2 Dataset profile

Table 6. Dataset after collection and preprocessing

Measure	Value
Raw records parsed	40,771
Clean records after normalisation	39,655
Rejected (by reason)	outside_window: 135, duplicate: 980, invalid_host: 1
Hosts resolved via DNS	1,555 of 2,316 non-platform domains
Distinct IP addresses	10,578
IPs geolocated	10,573 (100.0%)
Investigation window	30 days ending 2026-10-19

The five required data dimensions are all present. Entities: URLs, hosts, registered domains, IPs, hashes, ASNs, countries, malware families and reporters. Relationships: hosted_on, subdomain_of, resolves_to, in_asn, located_in, associated_with and reported_by. Timestamps: first and last seen. Locations: latitude/longitude, country and ASN. Threat indicators: URL, domain, IP:port and hash IOCs with threat type and confidence.

4. System Architecture

The platform is a Python pipeline with a Flask presentation layer. It is organised as eight layers, each in its own package under src/ and each testable on its own. Figure 1 shows the architecture; Table 7 maps each layer to its implementation.

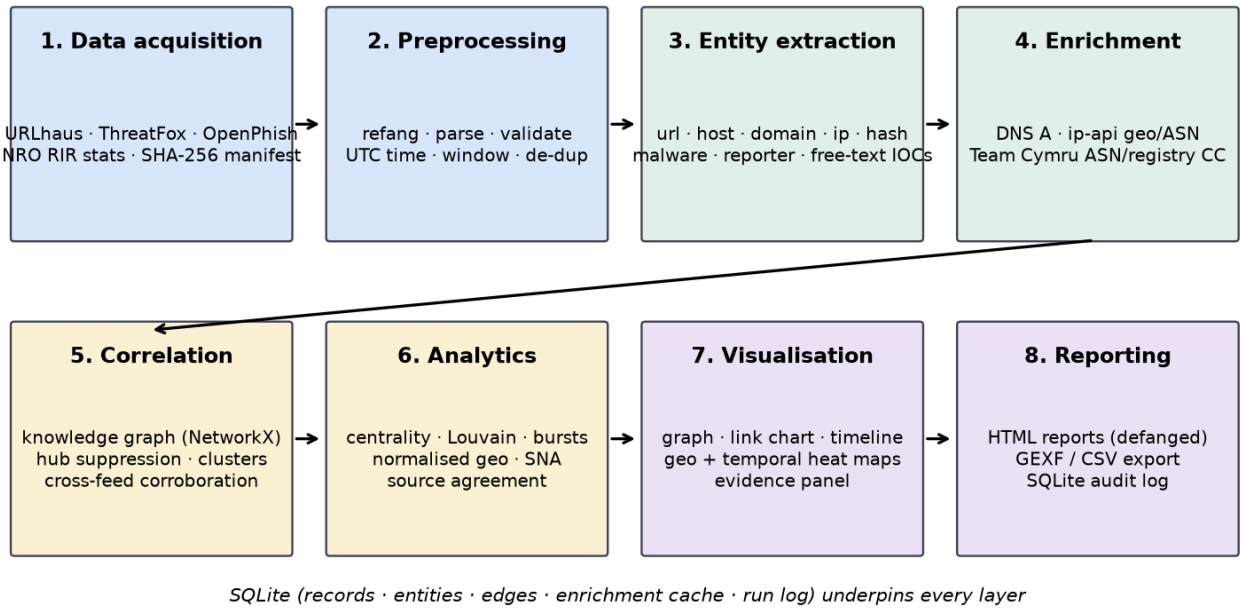


Figure 1. Layered architecture from data acquisition to reporting

Table 7. Architecture layers and implementation

Layer	Module	Responsibility	Key technique
Data acquisition	src/acquisition	Download feeds, manifest, snapshot, RIR stats	Streaming download, SHA-256, gzip snapshot
Preprocessing	src/preprocessing	Refang, parse, validate, UTC, window, de-dup	urllib, ipaddress, tldextract (bundled PSL)
Entity extraction	src/extraction	Typed entities + provenance edges; free-text IOC regex	Entity IDs '<type>:<value>' merge across feeds
Enrichment	src/enrichment	DNS, geolocation, two independent ASN sources	Threaded DNS, batch API, cache
Storage	src/storage	Records, entities, edges, cache, run log	SQLite
Correlation	src/correlation	Infrastructure graph, hub suppression, clusters, cross-feed	NetworkX connected components
Analytics	src/analytics	Centrality, Louvain, bursts, geo normalisation, SNA, agreement	NetworkX, pandas, z-scores
Visualization & reporting	src/web, src/reporting	Dashboard, evidence panel, HTML reports, GEXF/CSV	Flask, Cytoscape.js, Plotly, Leaflet, Jinja2

4.1 Data model

The knowledge graph uses typed nodes and typed, directed, provenance-labelled edges. An entity ID is '<type>:<value>' (for example ip:203.0.113.5), so an indicator reported by two feeds merges into one node automatically. That merge is the basis of cross-feed correlation. Every edge records the feed or method that created it, the source record ID, its time and its confidence. Feed attribution gets the ThreatFox confidence; tag-inferred family links get 50; DNS links get 90; geolocation links get 70. This is what lets the dashboard answer the question 'why are these two things connected?'.

Table 8. Relationship types in the knowledge graph

Relationship	From → To	Created by	Count
reported_by	indicator → reporter	feed record	39,114
hosted_on	url → host/ip	feed record	34,697

associated_with	indicator → malware	feed record	32,771
subdomain_of	host → domain	feed record	15,687
located_in	ip → country	ip-api	10,573
in_asn	ip → asn	ip-api	10,484
resolves_to	host → ip	DNS	2,137

5. Evaluation of Visualization Techniques

Each candidate technique was evaluated against the investigation questions in Section 1.2, using the evidence from the literature (Section 2) and from this dataset. Table 9 is the resulting suitability matrix (●●● = primary fit, ●● = supporting, ● = poor fit). Table 10 lists, for each technique, how it can mislead and the safeguard implemented against it.

Table 9. Suitability of visualization techniques per investigative question

Question	Graph (overview)	Link chart	Timeline	Geo heat map	Temporal heat map
IQ1 shared infrastructure	●●●	●●	●	●	●
IQ2 explain one indicator	●	●●●	●●	●	●
IQ3 start / peak / stop	●	●	●●●	●	●●
IQ4 where hosted	●	●	●	●●●	●
IQ5 tempo	●	●	●●	●	●●●
IQ6 brokers	●●●	●●	●	●	●
IQ7 reporting community	●●● (bipartite)	●●	●	●	●

Table 10. Strengths, failure modes and safeguards per technique

Technique	Best for	Weak for	How it misleads	Safeguard implemented
Graph (cluster overview)	Seeing groups, components, hubs; centrality	Large dense data (hairball) [L6]; reading time	Proximity read as relationship; hubs merge unrelated campaigns	Hub suppression; only infrastructure edges form clusters; cap of 600 nodes; size = betweenness
Link chart (ego expansion)	Explaining one indicator; pivoting step by step	Global patterns	Expanding through a CDN floods the chart; enrichment links look like feed facts	Per-relation expansion cap with hidden count; dashed lines for DNS/geo, dotted for tag-inferred; hub border; evidence panel
Timeline	Sequence, bursts, campaign duration	Relationships, place	Report time read as attack time; old IOCs re-listed look new	'Report time (UTC)' label; window filter; OpenPhish observation times excluded; z-score bursts
Geographic heat map / choropleth	Concentration in space	Causality, attribution	Raw counts mirror hosting size; one server with 10k URLs dominates; server location read as attacker location	Distinct-IP weighting; per-million-IPv4 normalisation; metric switch with warning; agreement rate shown
Temporal heat map (day × hour)	Periodicity, tempo	Exact sequence	Reporter working hours read as attacker hours	Caveat; UTC stated

6. Data Processing, Correlation and Analysis

6.1 Cleaning and normalisation

- **Refanging:** hxxp → http and [.] → . so defanged indicators match.
- **Parsing:** URLs are split into scheme, host, port and path; ThreatFox 'ip:port' into IP and port.

- **Validation:** hosts must be a valid IPv4/IPv6 address or a syntactically valid domain; invalid hosts are rejected and counted.
- **Registered domain:** a bundled public-suffix list gives deterministic registered domains (login.example.co.uk → example.co.uk) without a network call.
- **Time and de-duplication:** all timestamps are converted to ISO-8601 UTC; records outside the window are dropped; duplicates are dropped on (source, indicator).
- **Family harmonisation:** URLhaus tags are mapped to the ThreatFox family vocabulary (for example 'mirai' → Mirai), labelled 'tag-inferred' with lower confidence than explicit attribution.

6.2 Correlation and hub suppression

Correlation runs on an infrastructure graph containing only url, hash, host, domain and IP nodes, linked by hosted_on, subdomain_of and resolves_to edges. ASNs and countries never act as links, because one ASN can be a whole cloud provider. Before computing clusters, 637 hub nodes were suppressed. Hubs are known multi-tenant platforms (code hosting, CDNs, file sharing, dynamic DNS), plus any pivot whose two-hop neighbourhood spans more than 5 malware families. The connected components of what remains are candidate campaigns. The run found 5,019 clusters with at least two indicators.

Table 11. Highest-degree suppressed hubs

Suppressed hub	Reason	Connections
host:raw.githubusercontent.com	known multi-tenant platform	13,572
host:github.com	known multi-tenant platform	1,752
host:gitlab.com	known multi-tenant platform	693
ip:125.21.240.67	shared by 26 malware families	304
host:cdn.discordapp.com	known multi-tenant platform	150
domain:pages.dev	known multi-tenant platform	142
host:www.dropbox.com	known multi-tenant platform	87
host:drive.google.com	known multi-tenant platform	80
domain:workers.dev	known multi-tenant platform	59
domain:vercel.app	known multi-tenant platform	34

6.3 Graph analytics

On the hub-suppressed infrastructure graph (48,306 nodes, 20,155 edges, 28,470 components), degree centrality and approximate betweenness centrality (k = 400, fixed seed) were computed, together with Louvain communities (modularity 0.999). The very high modularity confirms what the components show: once hubs are removed, the infrastructure splits into many tight, separate groups rather than one giant network. Table 12 lists the most central nodes, which an analyst should pivot on first.

Table 12. Most connected infrastructure after hub suppression

Rank	Entity	Type	Degree centrality
1	176.65.134.121	ip	0.0076
2	91.92.242.236	ip	0.00228
3	31.56.209.86	ip	0.00199
4	45.153.34.235	ip	0.00143
5	176.65.139.135	ip	0.00118
6	154.83.85.63	ip	0.00108
7	176.65.139.206	ip	0.00099

8	130.12.180.212	ip	0.00097

6.4 Temporal analytics

Daily counts of newly reported indicators (mean 1240.6 per day) were tested for bursts with a z-score threshold of 2.0, both overall and per family.

Table 13. Strongest per-family bursts

Family	Day	Indicators	z-score
Stealc	2026-10-12	612	5.39
SmartLoader	2026-10-06	13,770	5.38
Unknown Loader	2026-10-18	220	5.32
Unattributed	2026-10-08	1,596	5.19
IClickFix	2026-10-18	138	4.4
Vidar	2026-10-18	311	4.29
IClickFix	2026-10-17	94	2.89
Vidar	2026-10-19	213	2.84
Mozi	2026-09-21	269	2.21
Mirai	2026-10-03	416	2.15

6.5 Geographic analytics

10,573 IPs were geolocated. Countries were measured three ways: URL count, distinct IP count, and distinct IPs per million allocated IPv4 addresses (NRO statistics). Only countries with at least 5 IPs were ranked by the normalised metric, so tiny denominators cannot dominate.

Table 14. Top hosting countries by distinct IPs

Country	Distinct IPs	URLs	IPs per million IPv4
China (CN)	4,955	7,970	14.45
United States (US)	915	1,515	0.57
Canada (CA)	639	2,209	9.42
Germany (DE)	400	1,289	3.16
India (IN)	378	830	9.0
Netherlands (NL)	372	2,257	7.76
Hong Kong (HK)	265	372	16.56
South Africa (ZA)	192	341	7.06
Russian Federation (RU)	189	343	4.18
Philippines (PH)	185	548	29.51

6.6 Social-network analysis of the reporting community

Reporters (pseudonymised) and malware families form a bipartite network. Projected onto reporters, two reporters are linked when they report the same family. 102 reporters cover 118 families. The top 5 reporters contribute 81.8% of the attributed indicators, and 64 families are reported by only one person. For an investigator this matters: a family seen by a single reporter has no independent corroboration in this dataset, and coverage gaps follow the interests of a few contributors.

7. Tool Implementation and Dashboard

The pipeline runs from one CLI (python main.py run-all, or the individual collect / enrich / analyze / report / serve steps). The Flask dashboard provides eight views over the same data. Each view states, on the page, what it does and does not show. Screenshots below were taken from the running platform on the analysed snapshot.

Table 15. Technology stack

Technology	Use
Python 3, pandas, NetworkX	Pipeline, analytics, graph algorithms
SQLite	Records, entities, edges, enrichment cache, run audit log
Flask + Jinja2	Dashboard and HTML reports
Cytoscape.js	Graph and link-chart rendering
Plotly.js	Timeline, choropleth, temporal heat map, ASN chart
Leaflet + leaflet.heat	Geographic heat map
dnspython, requests, tldextract, pycountry	Collection and enrichment
pytest; Git/GitHub (one commit per phase)	Testing and version control

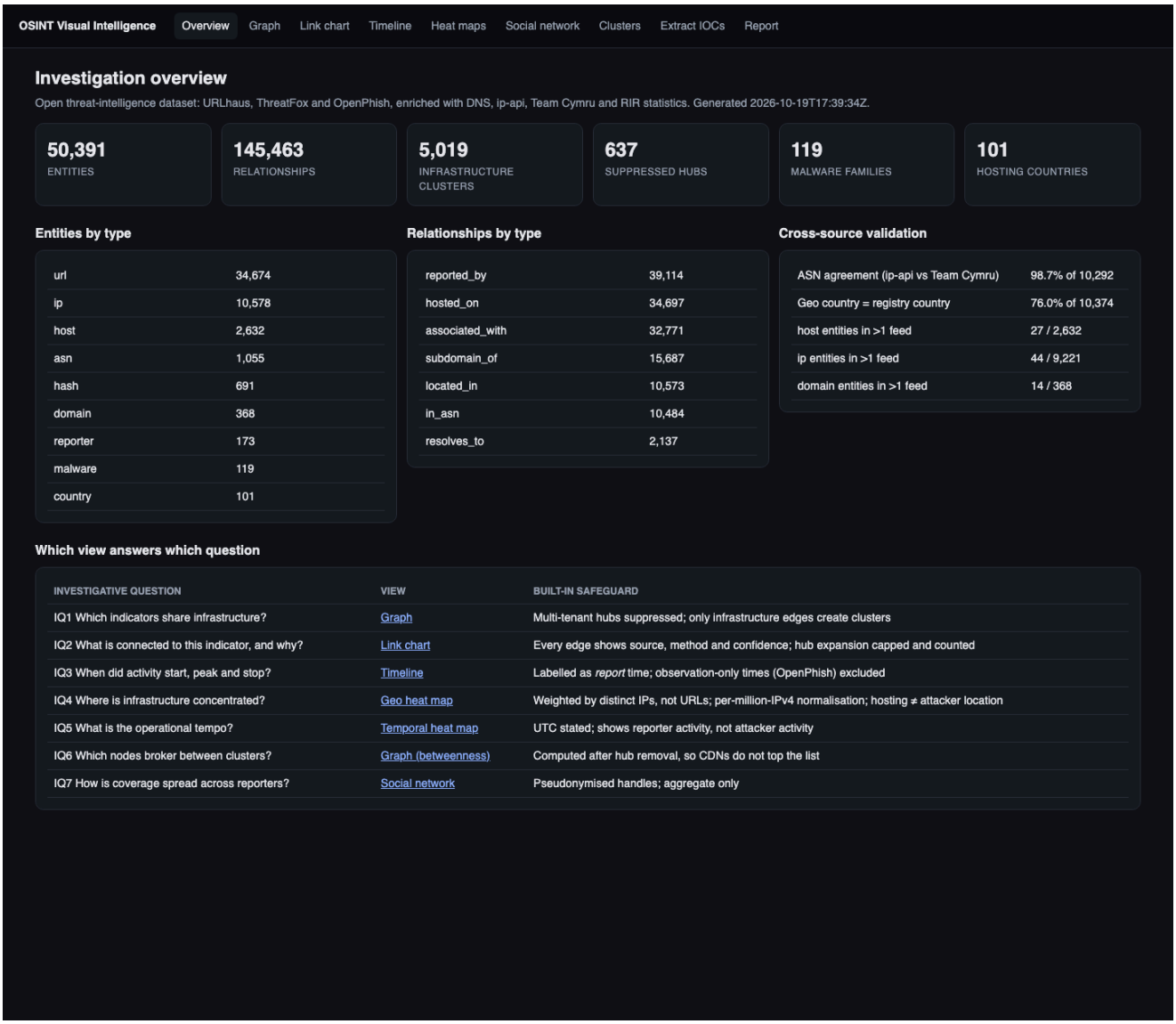


Figure 2. Overview: data volume, cross-source validation and the question-to-view guide

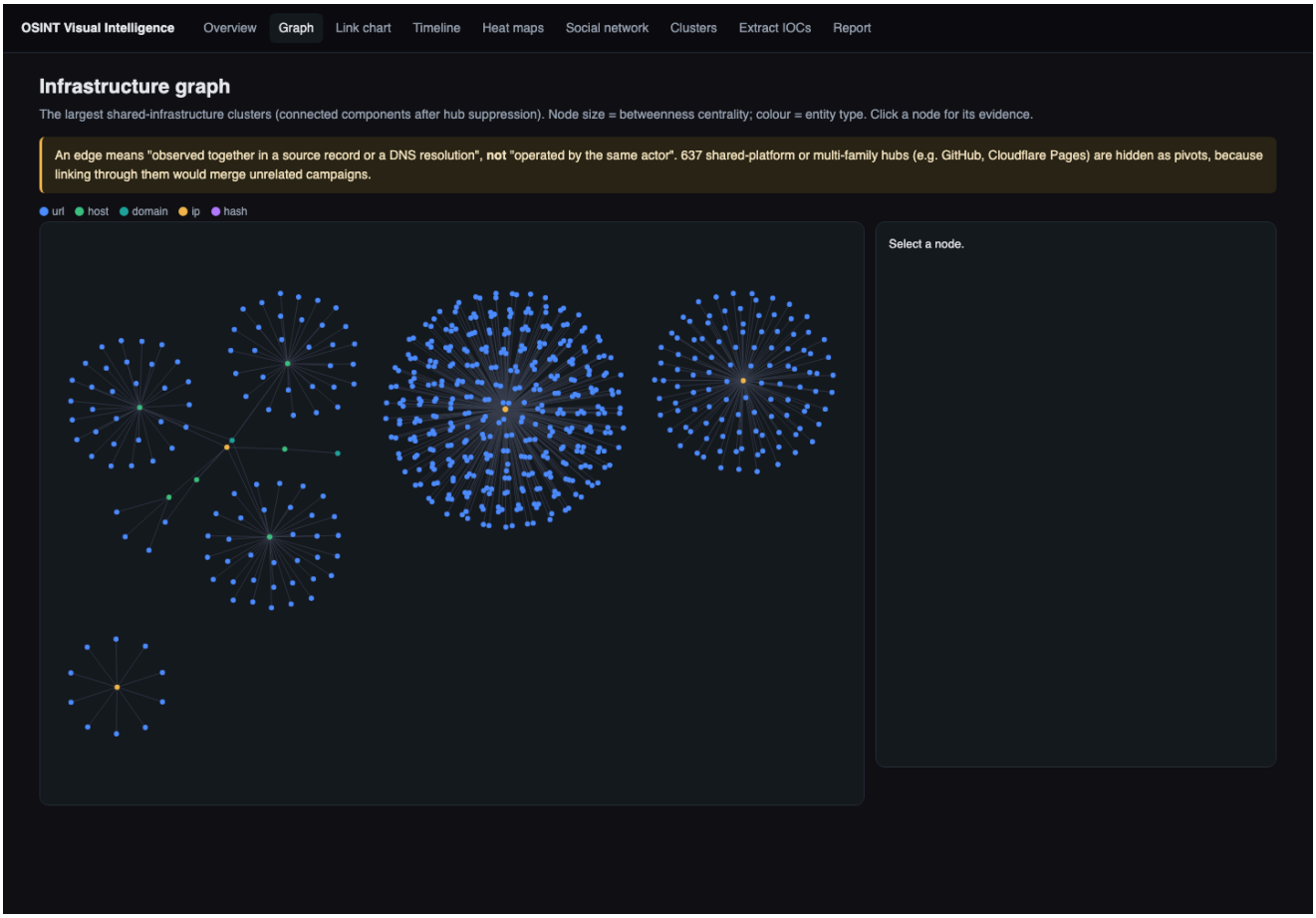


Figure 3. Infrastructure graph: largest clusters after hub suppression; node size = betweenness; the evidence panel shows a selected node's provenance

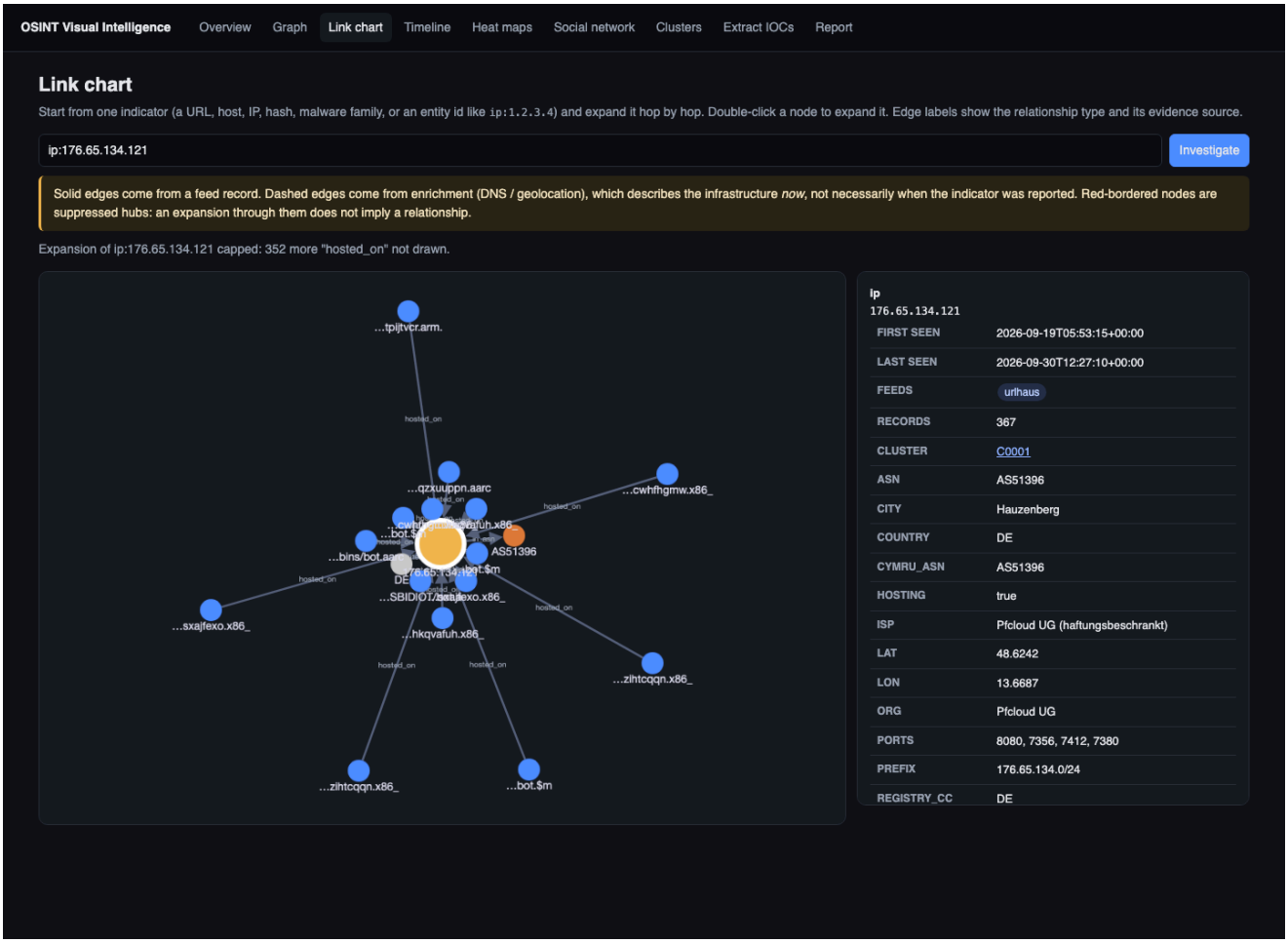


Figure 4. Link chart: expansion from one indicator; capped expansion with hidden-neighbour count; dashed enrichment edges

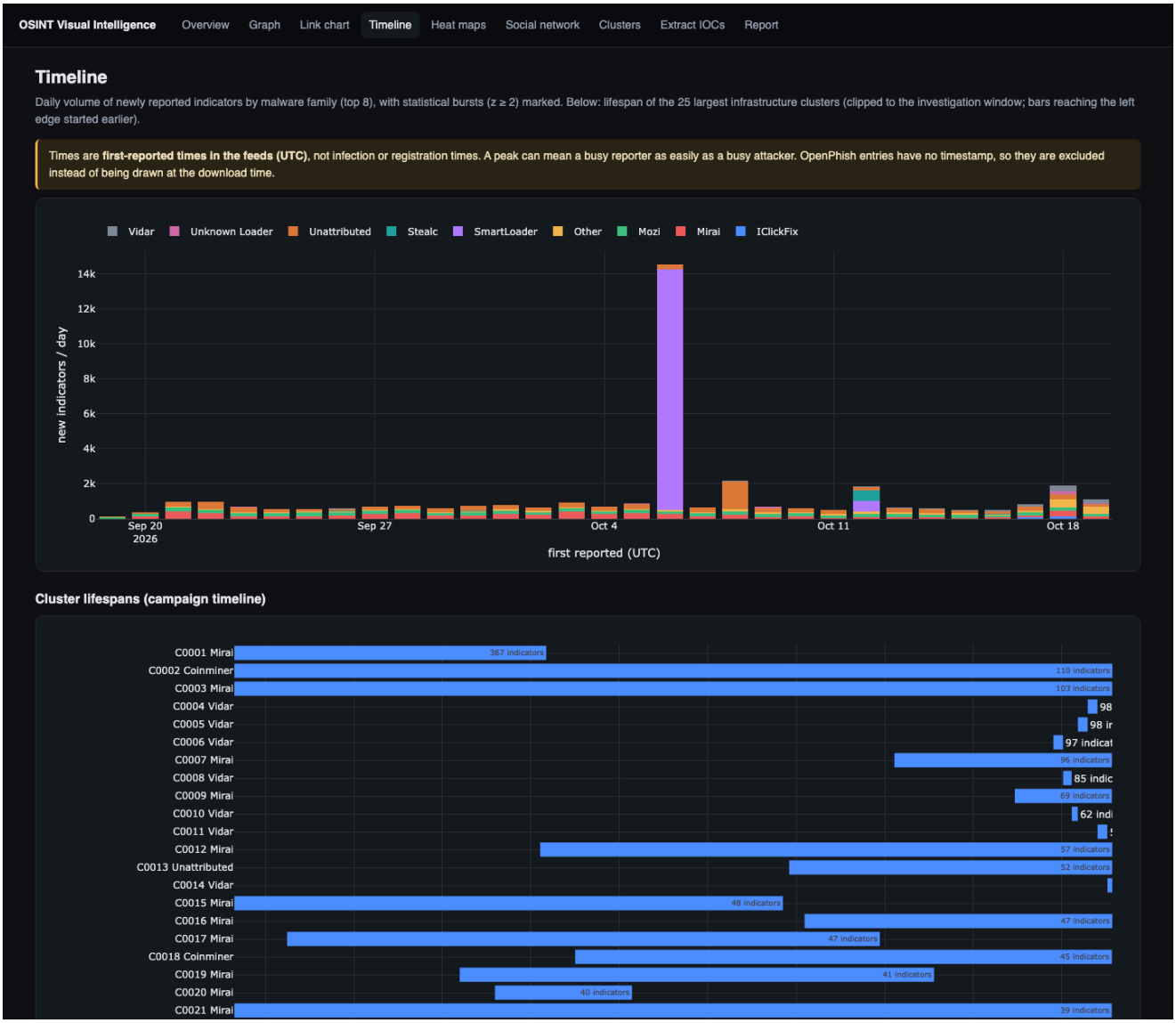


Figure 5. Timeline: daily new indicators by family with report-time caveat, and cluster lifespans

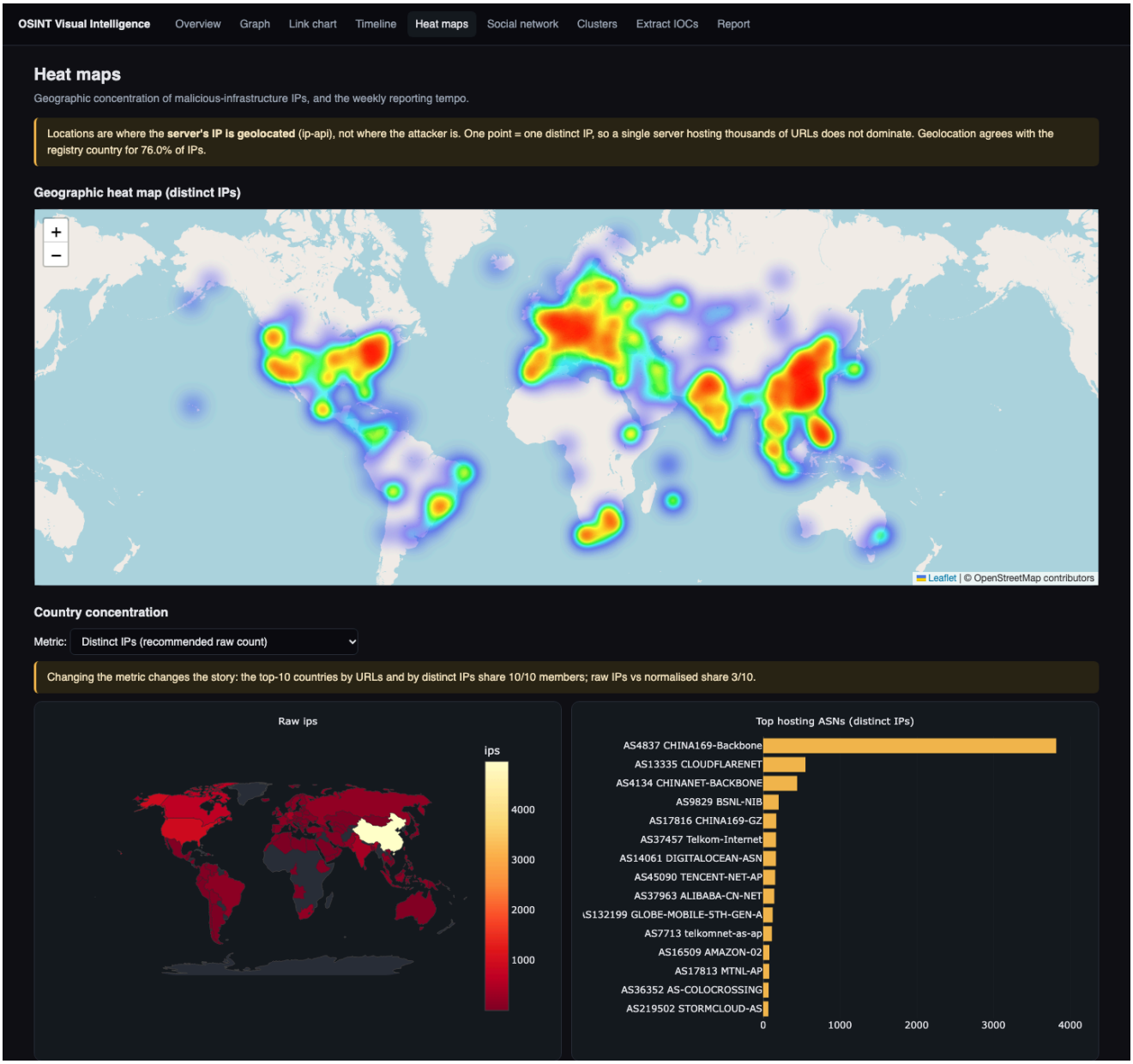


Figure 6. Geographic heat map weighted by distinct IPs, with country choropleth and hosting ASNs

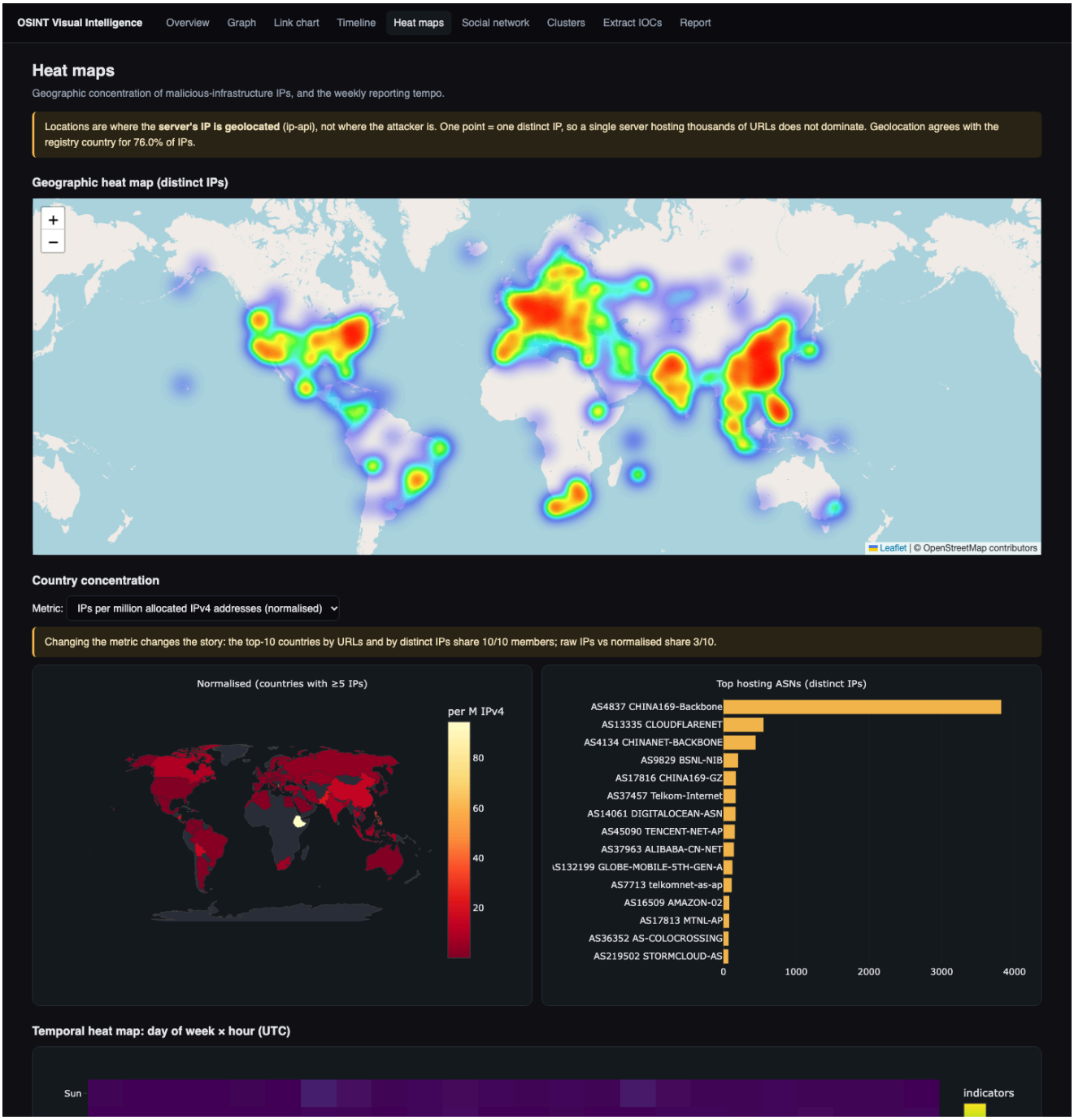


Figure 7. Choropleth after switching to per-million-IPv4 normalisation, and the day × hour temporal heat map

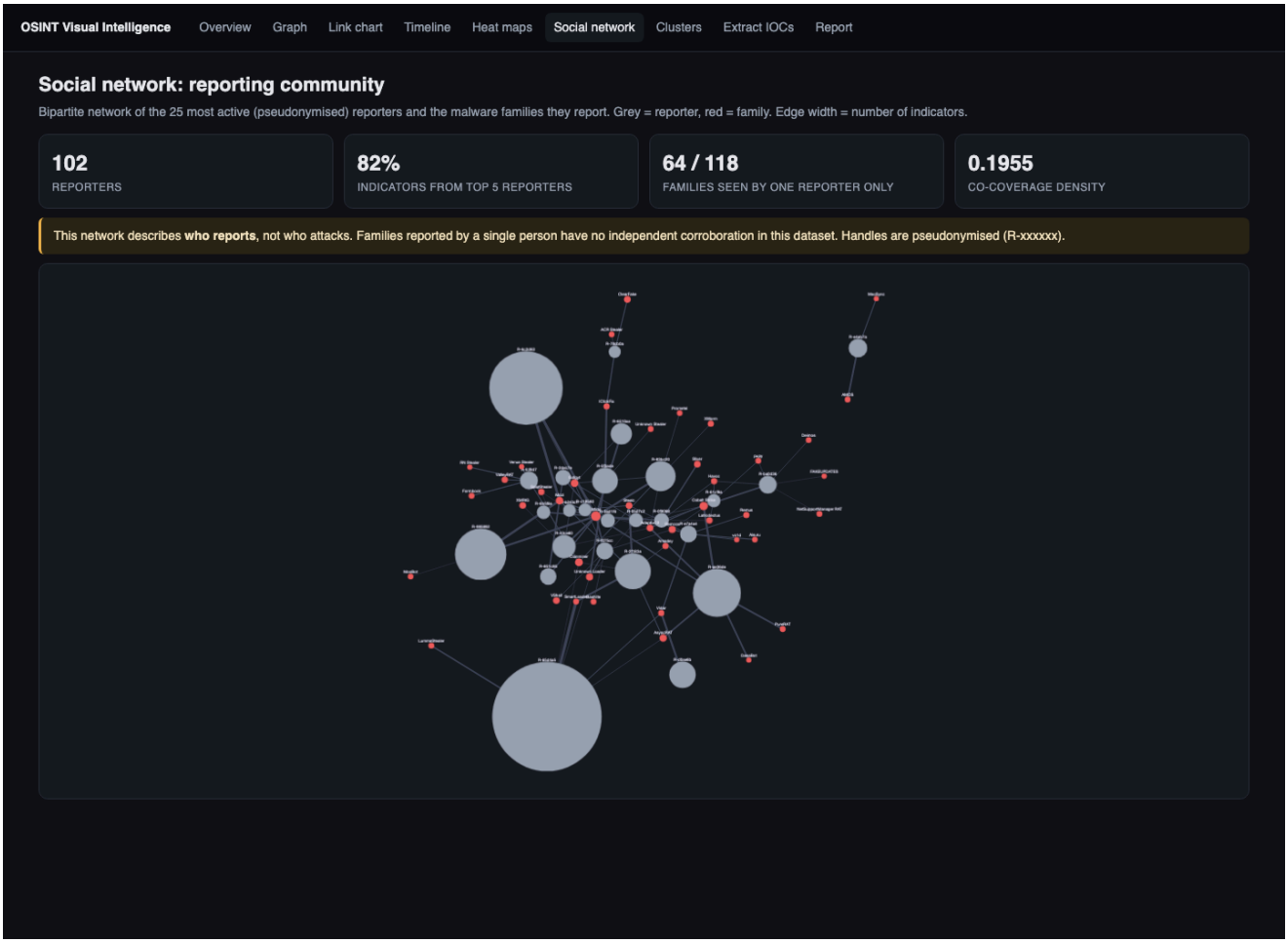


Figure 8. Social-network view of the (pseudonymised) reporting community

OSINT Visual Intelligence | Overview | Graph | Link chart | Timeline | Heat maps | Social network | Clusters | Extract IOCs | Report

Infrastructure clusters

Candidate campaigns: groups of indicators linked through shared hosts, registered domains or IPs (hubs excluded). Purity = share of family labels that agree with the dominant family. Export CSV · Export GEXF (Gephi)

ID	DOMINANT FAMILY	INDICATORS	IPS	PURITY	COUNTRIES	FEEDS	FIRST → LAST SEEN	
C0001	Mirai (180) Gafgyt (2) Bashlite (2)	367	1	0.978	DE	urlhaus	2026-09-19 → 2026-09-30	Report · Link chart
C0002	Coinminer (4) Amadey (2) Stealc (1)	110	1	0.5	NL	urlhaus, threatfox	2026-04-26 → 2026-10-19	Report · Link chart
C0003	Mirai (93) Coinminer (1) Cobalt Strike (1)	103	1	0.979	IN	urlhaus, threatfox	2022-12-30 → 2026-10-19	Report · Link chart
C0004	Vidar (96)	98	2	1.0	CA	threatfox	2026-10-18 → 2026-10-19	Report · Link chart
C0005	Vidar (96)	98	2	1.0	CA	threatfox	2026-10-18 → 2026-10-18	Report · Link chart
C0006	Vidar (95)	97	2	1.0	CA	threatfox	2026-10-17 → 2026-10-18	Report · Link chart
C0007	Mirai (17)	96	1	1.0	NL	urlhaus	2026-10-12 → 2026-10-19	Report · Link chart
C0008	Vidar (83)	85	2	1.0	CA	threatfox	2026-10-18 → 2026-10-18	Report · Link chart
C0009	Mirai (70)	69	1	1.0	NL	urlhaus, threatfox	2026-10-16 → 2026-10-19	Report · Link chart
C0010	Vidar (60)	62	2	1.0	CA	threatfox	2026-10-18 → 2026-10-18	Report · Link chart
C0011	Vidar (57)	59	2	1.0	CA	threatfox	2026-10-19 → 2026-10-19	Report · Link chart
C0012	Mirai (51)	57	1	1.0	NL	urlhaus	2026-09-30 → 2026-10-19	Report · Link chart
C0013		52	1	-	HK	urlhaus	2026-10-08 → 2026-10-19	Report · Link chart
C0014	Vidar (49)	51	2	1.0	CA	threatfox	2026-10-19 → 2026-10-19	Report · Link chart
C0015	Mirai (40)	48	1	1.0	NL	urlhaus	2026-09-19 → 2026-10-08	Report · Link chart
C0016	Mirai (2) Aisuru (1)	47	1	0.667	NL	urlhaus, threatfox	2026-10-09 → 2026-10-19	Report · Link chart
C0017	Mirai (41) Gafgyt (2)	47	1	0.953	NL	urlhaus	2026-09-21 → 2026-10-11	Report · Link chart
C0018	Coinminer (5) Mirai (3)	45	1	0.625	NL	urlhaus	2026-10-01 → 2026-10-19	Report · Link chart
C0019	Mirai (1)	41	1	1.0	NL	urlhaus	2026-09-27 → 2026-10-13	Report · Link chart
C0020	Mirai (34)	40	0	1.0	-	urlhaus	2026-09-28 → 2026-10-03	Report · Link chart
C0021	Mirai (35)	39	1	1.0	ZA	urlhaus, threatfox	2026-09-16 → 2026-10-19	Report · Link chart
C0022	Mirai (36)	39	1	1.0	NL	urlhaus	2026-09-28 → 2026-10-01	Report · Link chart
C0023	Mirai (33)	38	1	1.0	NL	urlhaus	2026-10-10 → 2026-10-19	Report · Link chart

Figure 9. Cluster table with purity, feeds, period and links to reports

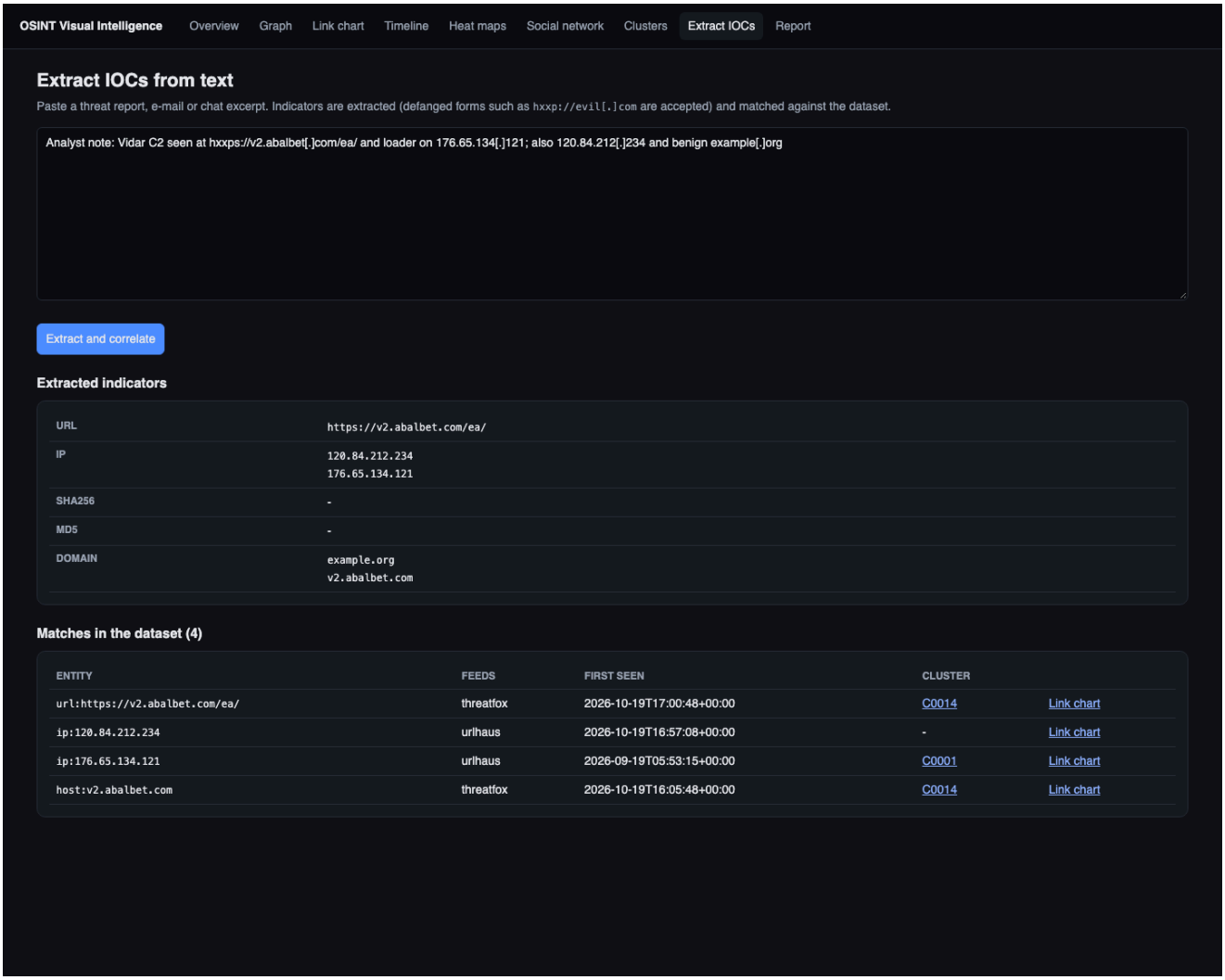
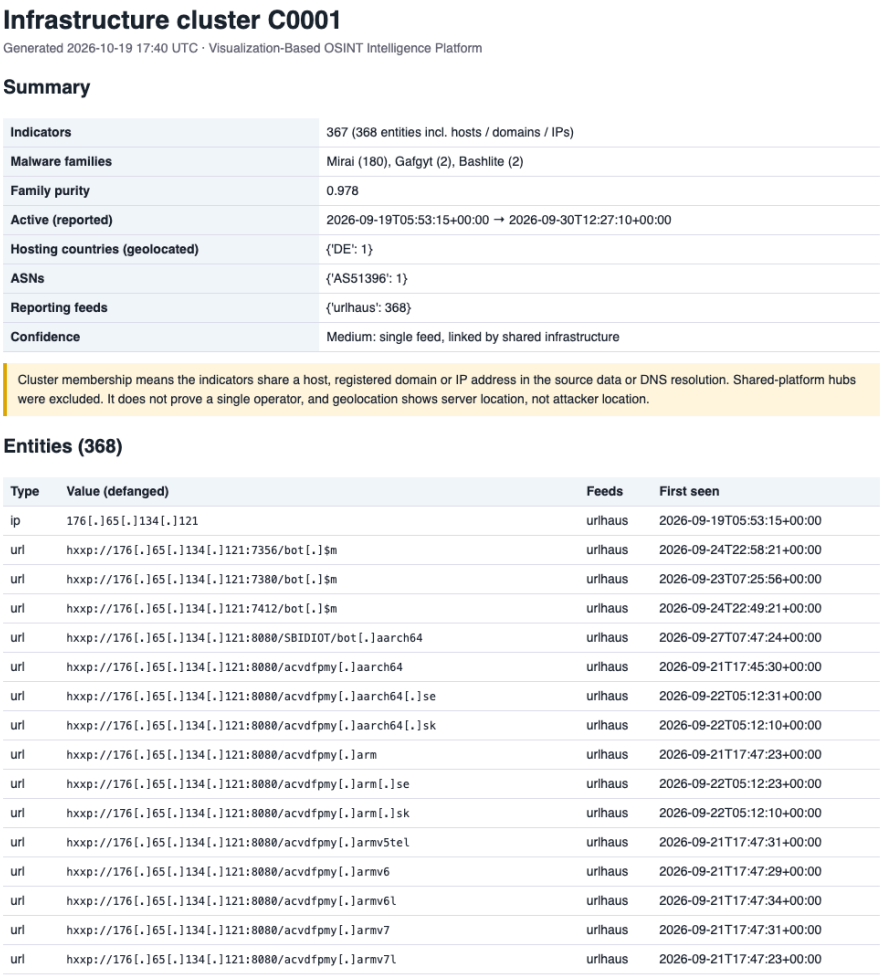


Figure 10. Free-text IOC extraction from a defanged report, correlated against the dataset



Infrastructure cluster C0001

Generated 2026-10-19 17:40 UTC · Visualization-Based OSINT Intelligence Platform

Summary

Indicators	367 (368 entities incl. hosts / domains / IPs)
Malware families	Mirai (180), Gafgyt (2), Bashlite (2)
Family purity	0.978
Active (reported)	2026-09-19T05:53:15+00:00 → 2026-09-30T12:27:10+00:00
Hosting countries (geolocated)	{'DE': 1}
ASNs	{'AS51396': 1}
Reporting feeds	{'urlhaus': 368}
Confidence	Medium: single feed, linked by shared infrastructure

Cluster membership means the indicators share a host, registered domain or IP address in the source data or DNS resolution. Shared-platform hubs were excluded. It does not prove a single operator, and geolocation shows server location, not attacker location.

Entities (368)

Type	Value (defanged)	Feeds	First seen
ip	176[.]65[.]134[.]121	urlhaus	2026-09-19T05:53:15+00:00
url	hxxp://176[.]65[.]134[.]121:7356/bot[.]$m	urlhaus	2026-09-24T22:58:21+00:00
url	hxxp://176[.]65[.]134[.]121:7380/bot[.]$m	urlhaus	2026-09-23T07:25:56+00:00
url	hxxp://176[.]65[.]134[.]121:7412/bot[.]$m	urlhaus	2026-09-24T22:49:21+00:00
url	hxxp://176[.]65[.]134[.]121:8080/SBIDIOT/bot[.]aarch64	urlhaus	2026-09-27T07:47:24+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]aarch64	urlhaus	2026-09-21T17:45:30+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]aarch64[.]se	urlhaus	2026-09-22T05:12:31+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]aarch64[.]sk	urlhaus	2026-09-22T05:12:10+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]arm	urlhaus	2026-09-21T17:47:23+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]arm[.]se	urlhaus	2026-09-22T05:12:23+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]arm[.]sk	urlhaus	2026-09-22T05:12:10+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]armv5tel	urlhaus	2026-09-21T17:47:31+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]armv6	urlhaus	2026-09-21T17:47:29+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]armv6l	urlhaus	2026-09-21T17:47:34+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]armv7	urlhaus	2026-09-21T17:47:31+00:00
url	hxxp://176[.]65[.]134[.]121:8080/acvdfpmy[.]armv7l	urlhaus	2026-09-21T17:47:23+00:00

Figure 11. Automatically generated, defanged cluster investigation report with evidence trail

8. Findings, Evidence and Validation

8.1 Key findings

- **Mass abuse of code hosting:** The largest single burst in the window was SmartLoader (13,770 new URLs on 2026-10-06, z = 5.38). Almost all of these are hosted on raw.githubusercontent.com. The timeline shows the burst, but only the graph view reveals that the URLs share no infrastructure beyond GitHub itself. Read alone, the timeline would suggest one massive campaign server.
- **Botnet distribution infrastructure:** The largest genuine cluster (C0001) holds 367 indicators dominated by Mirai (purity 0.978), on 1 IP address(es) in DE. This is a typical IoT-botnet distribution pattern: many short-lived URLs on a small set of servers.
- **Clusters are family-consistent:** Clusters where all labelled nodes carry one family dominate (weighted purity 0.924). Infrastructure linkage alone, with no malware labels used, recovers family-consistent groups, so the clusters are meaningful campaign candidates rather than artefacts.
- **Concentrated hosting:** The ten most heavily used hosting ASNs hold 5,955 of 10,573 geolocated IPs. 3,408 IPs are flagged as hosting/datacentre and 7,165 as residential or other, which is consistent with compromised home routers in IoT botnets.
- **Low feed overlap:** Only 44 of 9,221 IPs and 27 of 2,632 hosts appear in more than one feed. The feeds are largely complementary rather than redundant, which supports combining them. It also means most indicators rest on a single source.

8.2 Cross-source validation

Every derived attribute that could be checked against an independent source was checked.

Table 16. Cross-source validation results

Check	Sources compared	Compared	Agreement	Interpretation
ASN of each IP	ip-api vs Team Cymru (BGP)	10,292	98.7%	High: ASN attribution is reliable
Country of each IP	ip-api geolocation vs RIR registry country (Cymru)	10,374	76.0%	Country-level location is mostly reliable; disagreements are shown as a caveat on the map [L11]
Indicator corroboration	URLhaus vs ThreatFox vs OpenPhish	11,853 hosts+IPs	71 multi-feed	Multi-feed entities flagged as higher confidence in reports
Cluster consistency	Infrastructure clusters vs ThreatFox/URLhaus family labels	3,954	purity 0.924	Clusters align with independent family labels

Most frequent geolocation vs registry disagreements: CA vs US (584); DE vs US (104); NL vs DE (80); HK vs US (79); SG vs US (71). These are typically address blocks registered in one country and announced or used in another, which is exactly the uncertainty the map caveat warns about.

8.3 Misinterpretation experiments

Section 5 claims that each safeguard changes what an analyst would conclude. Three experiments test that claim on the real dataset.

Experiment A – hub suppression (graph). Without suppression the largest connected 'campaign' has 13,580 entities, glued together through platforms such as raw.githubusercontent.com. With suppression it has 368. An analyst working from the unsuppressed graph would have reported thousands of unrelated indicators as one operation.

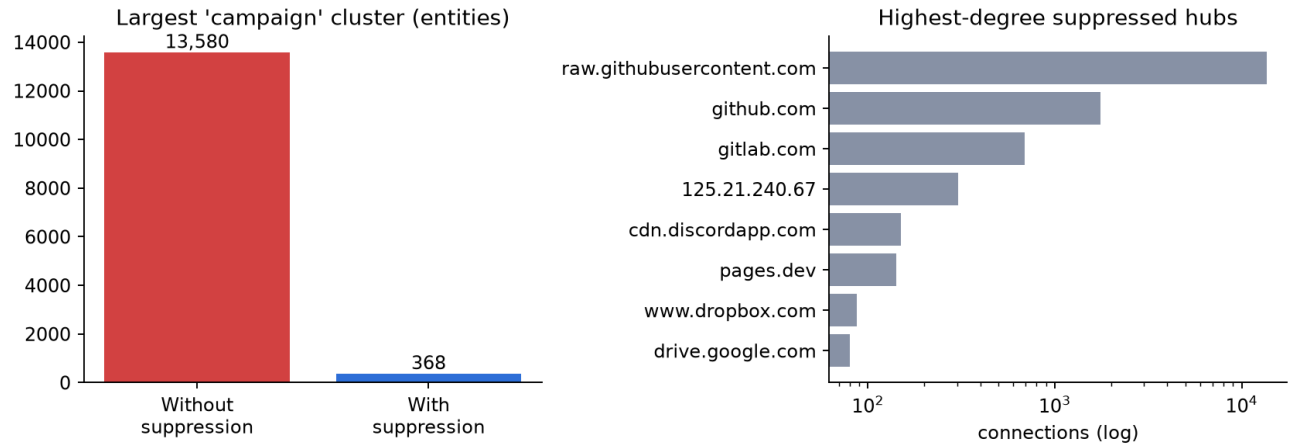


Figure 12. Experiment A: effect of hub suppression on the largest cluster, and the suppressed hubs with most connections

Table 17. Cluster structure with and without hub suppression

Metric	Without suppression	With suppression
clusters	5,036	5,019
largest cluster	13,580	368
labelled clusters	3,968	3,954
mixed family clusters	557	548
weighted purity	0.933	0.924

Purity is slightly lower with suppression. Removing hubs splits the giant component into many small clusters, so small mixed clusters now count individually, while the giant cluster previously counted as one mostly-SmartLoader group. Purity alone would therefore have rewarded the false merge. This is why cluster size is reported together with purity.

Experiment B – metric choice (heat map). Ranking countries by URLs, by distinct IPs and by IPs per million IPv4 gives different answers. The top 10 by URLs and by distinct IPs share 10 countries (Spearman ρ = 0.954); distinct IPs and normalised share 3 (ρ = 0.362) across 68 countries. URLs and IPs agree on membership but not on order: the Netherlands is 2nd by URLs but 6th by distinct IPs, because a few Dutch servers host many URLs each. Normalisation changes the picture completely: the United States holds the second-most IPs but drops out of the top 10 relative to its very large address space, while smaller countries rise. The raw map answers 'where is most infrastructure', the normalised map answers 'which countries are over-represented for their size'. The dashboard keeps both, labelled, and warns that URL counts are inflated.

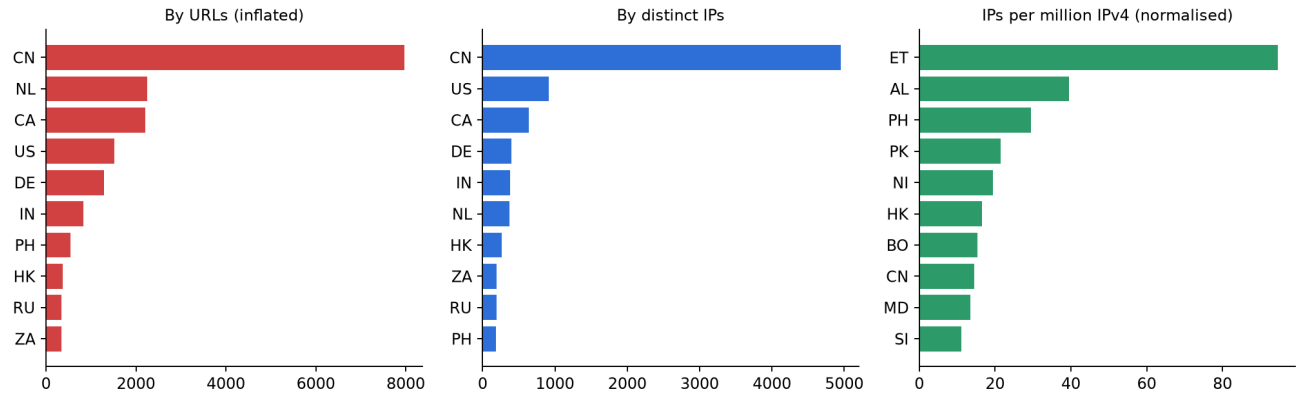


Figure 13. Experiment B: top-10 hosting countries under three metrics

Experiment C – time semantics (timeline). ThreatFox's 'recent' export re-lists indicators whose first sighting dates back to 2021–2022 because they were seen again. Plotting first-seen dates naively produced a false Cobalt Strike 'burst' in December 2022 during development. Restricting the timeline to the investigation window and labelling the axis as report time removed it. OpenPhish has no timestamps at all; drawing its 300 URLs at the download time would have created another artificial spike, so they are excluded from the timeline.

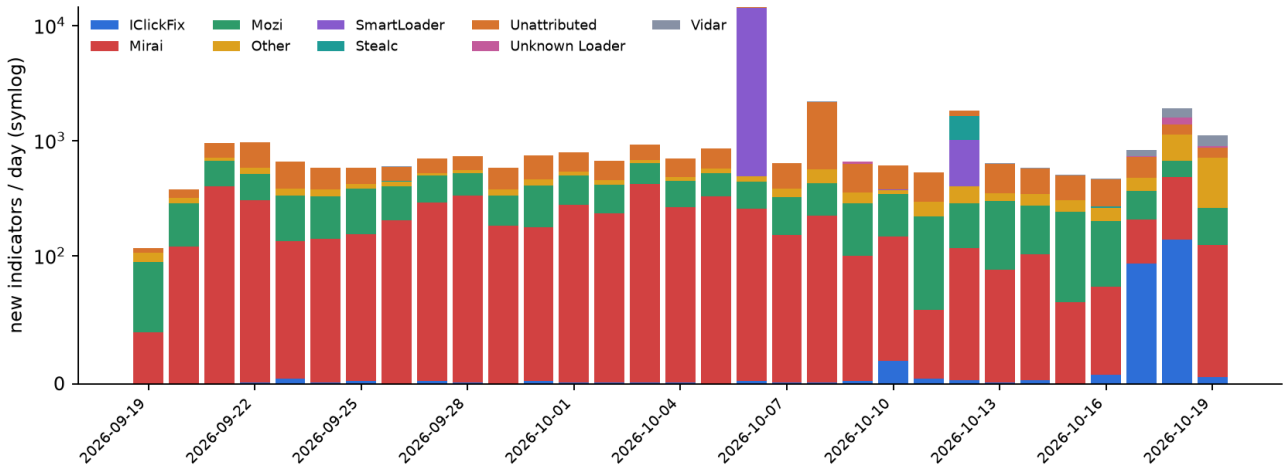


Figure 14. Daily newly reported indicators by family (symmetric-log scale)

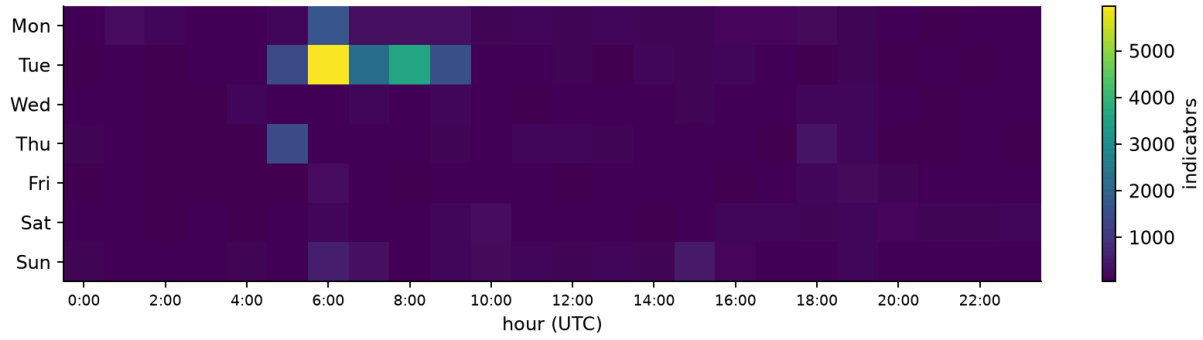


Figure 15. Temporal heat map: reports by weekday and hour (UTC)

8.4 Functional testing

A pytest suite (tests/test_pipeline.py, 10 tests, all passing) covers refang/defang, URL and ip:port parsing, registered-domain extraction, rejection of bad timestamps and duplicates, defanged free-text IOC extraction, provenance on extracted edges, reporter pseudonymisation, and both hub-suppression rules on a synthetic graph in which two unrelated families share one host. Every dashboard route and API was smoke-tested through Flask's test client. The offline mode was verified by restoring the snapshot and checking all SHA-256 hashes.

8.5 Limitations

- **Reporting bias:** Feeds reflect what volunteers report. Coverage follows reporter interest (Section 6.6), and absence from a feed is not evidence of safety.
- **Enrichment time-lag:** DNS and geolocation describe infrastructure at enrichment time, not at report time. Fast-flux hosts may have moved. Enrichment edges are therefore drawn dashed.
- **Geolocation accuracy:** ip-api is a single commercial geolocation database. It was cross-checked against registry country, but registry country is itself only an approximation of physical location.
- **Family labels:** mapping URLhaus tags to families is heuristic and is labelled 'tag-inferred'. Many indicators remain unattributed.
- **Hub rules:** the shared-platform list is curated by hand, and the family threshold (5) is a tunable parameter. Some legitimately shared criminal bulletproof hosting may be suppressed, and some unknown platforms may be missed.
- **Service limits:** the free ip-api tier is rate-limited (about 15 requests per minute) and intended for non-commercial use. A production deployment would need a licensed provider.
- **No user study:** the safeguards were evaluated through data experiments, not a user study with analysts.

9. Ethical and Legal Considerations

- **Passive collection only:** all data comes from public feeds and passive lookups (DNS, registry, geolocation). The platform never connects to, scans, downloads from or logs in to any indicator.
- **Terms of service:** abuse.ch data is published for free use under its terms of use; OpenPhish Community and ip-api free tiers are used for non-commercial, academic purposes within documented rate limits; all requests carry an identifying User-Agent.
- **Privacy and minimisation:** the platform analyses infrastructure, not people. Reporter handles are replaced by stable pseudonyms (R-xxxxxx) in every output and used only in aggregate. No WHOIS registrant data or other personal data is collected, in line with purpose limitation and data minimisation [L4], [R5].
- **Safe handling of malicious content:** the committed snapshot is gzip-compressed, so live malicious URLs are not clickable or indexed in the repository. HTML reports defang every indicator (hxxp, [.]).

- **Responsible presentation:** every view states what it does not show (hosting location ≠ attacker, report time ≠ attack time, link ≠ same operator). Cluster reports state their confidence and evidence. Results are triage aids to be verified by an analyst before any blocking or reporting [R7].
- **Legal:** no system is accessed without authorisation, which keeps the work clear of computer-misuse provisions such as the Information Technology Act, 2000 (India) [R6] and the US Computer Fraud and Abuse Act [R8]. This is a design intention, not legal advice.

10. Impact and Significance

- **Integration:** one pipeline replaces manual cross-referencing of three feeds and four enrichment services, producing a 50,391-entity knowledge graph in minutes on a laptop.
- **Question-driven visualization:** each investigative question is matched to the visualization that answers it best, with a written justification and failure modes.
- **Measurably fewer misreadings:** the experiments show that ignoring shared platforms, metric choice or time semantics would change the investigation's conclusions. The safeguards are therefore functional, not cosmetic.
- **Evidence-grade outputs:** every edge is traceable to a feed record or enrichment method, and every number in this report can be reproduced offline from a hash-verified snapshot.
- **Interoperability:** GEXF and CSV exports let analysts continue the work in Gephi or a spreadsheet.

11. Future Work

- Passive DNS history so that host → IP links are time-stamped at report time, removing the enrichment time-lag.
- A second geolocation provider and latency-based checks to quantify geolocation uncertainty per IP.
- Learning the hub threshold from data (for example the tenant count per IP from passive DNS) instead of a fixed list.
- A controlled user study with analysts comparing task accuracy with and without the safeguards.
- STIX 2.1 export and MISP integration for sharing clusters.
- Scheduled collection to build a longitudinal dataset and track clusters over months.

12. Conclusion

The project delivered a working, reproducible visualization-based OSINT intelligence platform. It turns three public feeds and four enrichment sources into a provenance-labelled knowledge graph of 50,391 entities and 145,463 relationships, analyses it with graph, temporal, geographic and social-network methods, and presents it through a graph, a link chart, a timeline and heat maps. The evaluation showed that no single technique answers every investigative question: graphs suit structure, link charts suit explanation, timelines suit sequence and heat maps suit concentration. Each also has a characteristic way of misleading. Building safeguards into the views (hub suppression, distinct-IP weighting and normalisation, report-time semantics, and evidence on every edge) changed the conclusions an analyst would draw from the same data. In particular, it prevented a false 13,580-entity 'campaign'. The findings were cross-validated against independent sources, and the limitations are stated explicitly.

Repository

Source code, phase-by-phase commit history, the frozen dataset snapshot and setup instructions: https://github.com/Taniaa-p/OSINT-2

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